Dimensionality as Cognitive Capacity: An Interpretive Framework for Multiscale Neural Structure

Mark Rowe Traver
Sentient-Field Initiative (Braintrust)

April 2026

Abstract

Multiscale dimensionality profiling has emerged as a powerful technique for characterizing neural representations across spatial scales. However, the relationship between the parameters extracted from dimensionality profiles and cognitive function remains undefined. We introduce an interpretive framework that maps multiscale dimensionality parameters to cognitively meaningful variables. Specifically, we propose that amplitude reflects representational capacity, midpoint reflects organizational integration scale, and steepness reflects cross-scale integration efficiency. This framework enables testable predictions about how cognitive manipulations should affect dimensionality parameters. We outline empirical test pathways using fMRI, EEG/MEG, and behavioral paradigms. This work defines a formal mapping between multiscale dimensionality structure and interpretable cognitive variables, enabling hypothesis-driven applications of dimensionality analysis in neuroscience.

1 Introduction

Multiscale dimensionality analysis provides a framework for characterizing the effective dimensionality of neural representations as a function of neighborhood scale k [4, 2]. The method quantifies how information about local neighborhood structure accumulates across spatial scales, from local neighborhoods to global patterns [1, 6]. Applications in neuroscience have demonstrated that neural population activity exhibits characteristic dimensionality profiles that vary across brain areas, behavioral states, and experimental conditions [3, 5, 7].

A fundamental limitation of current dimensionality analysis approaches is that while dimensionality can be precisely measured, the relationship between dimensionality parameters and cognitive function remains undefined. Raw dimensionality values lack cognitive interpretation—knowing that a neural population has effective dimensionality $D = 50$ provides no insight into what cognitive functions that dimensionality supports or how it relates to perception, memory, or action. This interpretability gap limits the utility of dimensionality analysis for hypothesis-driven neuroscience research.

Previous work has established that multiscale dimensionality profiles follow characteristic functional forms. Theoretical analysis demonstrates that residual dimensionality profiles—following null-model subtraction—admit low-dimensional parameterization in terms of sigmoid functions 5. The three-parameter sigmoid form captures amplitude, midpoint, and steepness. However, the cognitive meaning of these parameters has not been established.

This paper introduces a formal interpretive framework that maps multiscale dimensionality parameters to cognitively meaningful variables. We propose mechanistic interpretations for each parameter and derive testable predictions about how cognitive manipulations should affect dimensionality structure. This framework transforms dimensionality analysis from a descriptive technique into a hypothesis-generating and hypothesis-testing tool for cognitive neuroscience.

We define a formal mapping between multiscale dimensionality structure and interpretable cognitive variables.

2 Mathematical Framework

2.1 Definitions

We define dimensionality in terms of representational degrees of freedom:

$$\dim(\mathbf{X}) = \operatorname{rank}(\mathbf{\Sigma_X}) \tag{1}$$

where $\mathbf{X} \in \mathbb{R}^{N \times D}$ is a data matrix with N samples and D dimensions, and $\mathbf{\Sigma_X} = \operatorname{Cov}(\mathbf{X})$ is the covariance matrix.

This definition reflects the number of independent axes of variation in the representation. In neural systems, this corresponds to the number of independent patterns of population activity available for encoding information.

Residual dimensionality profiles are computed by subtracting null-model predictions:

$$D_{\text{res}}(k) = D_{\text{obs}}(k) - D_{\text{null}}(k) \tag{2}$$

where $D_{\text{obs}}(k)$ is the observed dimensionality at scale k, and $D_{\text{null}}(k)$ is the expected dimensionality under a null model of random structure 3. The residual profile isolates system-specific structure from generic statistical effects.

The sigmoid parameterization of residual dimensionality profiles takes the form:

$$D(k) = \frac{A}{1 + e^{-\beta(k-k_0)}} \tag{3}$$

where A is the amplitude, k_0 is the midpoint, and β is the steepness. Each parameter has both mathematical definition and cognitive interpretation, as developed below.

The null model prediction $D_{\text{null}}(k)$ is derived from analytical expectations for the eigenvalue spectrum of isotropic Gaussian data under random matrix theory (Marchenko–Pastur distribution),

implemented as a matched-dimensional surrogate with identical sample size and variance structure 1967distribution.

3 Interpretive Mapping

We now introduce the interpretive framework that maps sigmoid parameters to cognitively meaningful variables.

3.1 Amplitude and Representational Capacity

The amplitude parameter A is defined mathematically as the saturation level of the sigmoid function:

$$A = \lim_{k \to \infty} D(k) \tag{4}$$

Conceptually, A reflects the total accessible dimensional structure revealed across neighborhood scales. We propose that A corresponds to the representational capacity of the system.

Interpretive definition: Amplitude A is the maximum dimensionality accessible to the system across all scales, corresponding to the total pool of representational degrees of freedom available for encoding information under the current behavioral or cognitive state.

In neural terms, amplitude reflects the range of distinct neural activity patterns the system can express. Higher amplitude indicates a greater capacity for diverse representational states, enabling more complex encoding of stimuli, actions, or cognitive content.

This interpretation is grounded in the relationship between dimensionality and information capacity. The number of independent dimensions available for representation sets an upper bound on the amount of distinguishable states that can be encoded. Amplitude measures this bound empirically.

Implications:

- Cognitive load should correlate with amplitude
- Expertise should correlate with amplitude (more available states for skilled behavior)
- Pathologies affecting neural computation should reduce amplitude

3.2 Midpoint and Organizational Integration Scale

The midpoint parameter k_0 is defined mathematically as the inflection point of the sigmoid:

$$k_0 = \arg\min_{k} |D(k) - A/2| \tag{5}$$

Conceptually, k_0 reflects the neighborhood scale at which half the total dimensional structure is revealed. We propose that k_0 corresponds to the characteristic scale of organizational integration.

Interpretive definition: Midpoint k_0 is the spatial scale at which half the representational capacity becomes accessible, indicating the characteristic scale of integration in the representation.

In neural terms, k_0 reflects the spatial scale at which neural integration operates. Lower k_0 indicates integration at smaller spatial scales (more local processing), while higher k_0 indicates integration across larger spatial extents (more global processing).

This interpretation connects to established neuroscience concepts of local versus global processing. Cortical microcircuits operate at small spatial scales, while large-scale networks integrate information across brain regions.

Implications:

- Focused attention should decrease k_0 (more local processing)
- Global integration demands should increase k_0 (more global processing)
- Hierarchical processing should be reflected in k_0 gradients across areas

3.3 Steepness and Integration Efficiency

The steepness parameter β is defined mathematically as the derivative magnitude at the inflection point:

$$\beta = \left| \frac{dD}{dk} \right|_{k=k_0} \tag{6}$$

Conceptually, β reflects the rate at which dimensionality accumulates across scales. We propose that β corresponds to the efficiency of cross-scale integration.

Interpretive definition: Steepness β is the rate of dimensionality accumulation across neighborhood scales, indicating how efficiently the representation integrates information across spatial scales.

In neural terms, higher β indicates faster transition from local to global structure, reflecting more efficient integration. Lower β indicates more gradual accumulation, reflecting distributed processing across scales.

This interpretation connects to the efficiency of neural information processing. Faster integration across scales (higher β) could reflect more efficient communication between processing stages.

Implications:

- Expertise should increase β (more efficient processing)
- Learning should increase β (improved integration efficiency)
- Neural disorders affecting connectivity should reduce β

4 Testable Predictions

We derive five testable predictions from the interpretive framework.

4.1 Prediction 1: Task Complexity and Amplitude

Prediction: Increasing task complexity will increase amplitude A.

Rationale: More complex cognitive tasks require more diverse representational states to encode the increased informational content. The representational capacity must expand to accommodate additional distinguishable states.

Test:

1. Acquire fMRI or electrophysiology data during tasks of varying complexity (e.g., n-back tasks with increasing load, problem-solving tasks of varying difficulty)
2. Compute dimensionality profiles at each complexity level
3. Fit sigmoid functions and extract amplitude A
4. Expected: Positive correlation between task complexity and A

Measurement: Dimensionality computed using k-NN variance method; complexity operationalized as number of distinct stimuli, response options, or working memory load.

4.2 Prediction 2: Noise and Parameters

Prediction: Increased noise will decrease amplitude A and steepness β.

Rationale: Noise degrades the precision of neural representations, reducing the number of reliable distinguishable states and slowing integration across scales.

Test:

1. Manipulate signal-to-noise ratio in neural recordings (e.g., add visual noise, acoustic noise, or collect data in varying SNR conditions)
2. Compute dimensionality profiles under each noise level
3. Fit sigmoid functions and extract parameters
4. Expected: Negative correlation between noise and both A and β

Measurement: SNR quantified from recording characteristics or estimated from signal variability.

4.3 Prediction 3: Learning and Midpoint

Prediction: Learning/training will shift midpoint k_0 toward optimal integration scale.

Rationale: As a system learns a task, the integration scale should adapt to the demands of the task. Expert performers should show integration at scales optimal for the task.

Test:

1. Acquire neural data during task learning over multiple sessions

2. Compute dimensionality profiles at each learning stage

3. Expected: Systematic shift in k_0 toward task-optimal scale with learning

Measurement: Learning quantified by task performance; k_0 computed from fitted sigmoid functions.

4.4 Prediction 4: Attention and Steepness

Prediction: Focused attention will increase steepness β.

Rationale: Attention enhances processing efficiency, leading to faster integration across scales.

Test:

1. Acquire neural data during attention and divided-attention conditions

2. Compare steepness β between conditions

3. Expected: Higher β during focused attention

Measurement: Attention manipulated via task instructions; β from fitted sigmoid functions.

4.5 Prediction 5: Pathology and Amplitude

Prediction: Neural pathologies affecting representational capacity will reduce amplitude A.

Rationale: Pathologies that degrade neural population structure (e.g., neurodegeneration, stroke, developmental disorders) should reduce the pool of available representational states.

Test:

1. Compare dimensionality profiles in clinical populations versus controls

2. Compare amplitude A between groups

3. Expected: Reduced A in affected populations

Measurement: Clinical populations identified via diagnostic criteria; A from fitted sigmoid functions.

5 Empirical Test Pathways

We describe empirical approaches for testing the predictions outlined above.

5.1 fMRI Paradigms

Task Complexity Design:

- Working memory n-back task (1-back, 2-back, 3-back conditions)

- Compute voxel-wise dimensionality profiles across conditions
- Extract amplitude for each complexity level
- Expected: Positive correlation with load

Data requirements: High-resolution fMRI (>= 3T), whole-brain coverage, sufficient trials per condition.

Analysis: Motion correction, spatial normalization, temporal filtering; k-NN dimensionality computation at multiple scales; sigmoid fitting.

Outcome confirmation: Positive correlation between task load and amplitude with $p < 0.05$.

5.2 EEG/MEG Paradigms

Attention Design:

- Visual oddball task with attend and ignore conditions
- Compute sensor-level dimensionality profiles
- Compare steepness β between conditions
- Expected: Higher β during attend condition

Data requirements: High-density EEG (>= 64 channels) or MEG (>= 100 channels), sufficient trials.

Analysis: Preprocessing following standard pipelines; dimensionality computation in sensor space or source space.

Outcome confirmation: Significant increase in β during attention with $p < 0.05$.

5.3 Behavioral Paradigms

Learning Design:

- Acquire neural data during multiple learning sessions
- Track changes in midpoint k_0 over learning
- Expected: Systematic shift in k_0 with learning

Data requirements: Longitudinal recording sessions, consistent task across sessions.

Analysis: Compare k_0 across learning stages; correlation with behavioral performance.

Outcome confirmation: Significant correlation between k_0 and learning stage.

6 Relation to SFH-SGP Framework

We situate this interpretive framework within the broader context of dimensionality analysis.

We interpret dimensionality as reflecting the degrees of interaction available within the neural substrate. In sentient field terms, dimensionality quantifies the accessible interaction space available for representing cognitive content. Higher dimensionality corresponds to more degrees of interaction, enabling richer representational structure.

Residual dimensionality, following null-model subtraction, isolates the system-specific structure from generic statistical effects. This residual structure reflects the particular organizational principles of the system—the specific way the neural substrate is configured to represent information.

The sigmoid parameterization provides a compact description of residual structure. The three parameters (amplitude, midpoint, steepness) capture distinct aspects of system organization. This mapping enables interpretation of dimensionality in terms of cognitively meaningful variables.

This framework is consistent with and extends existing dimensionality analysis approaches. We build on the technical foundation established in prior work while adding interpretability.

The mapping between parameters and cognitive variables enables hypothesis-driven applications. Researchers can now generate predictions about how cognitive manipulations should affect dimensionality parameters, test those predictions empirically, and refine the interpretive framework based on results.

7 Discussion

This paper introduces an interpretive framework that transforms dimensionality analysis from a descriptive technique into a hypothesis-testing tool. By mapping sigmoid parameters to cognitively meaningful variables, we enable researchers to generate and test specific predictions about the relationship between neural structure and cognitive function.

The value of this framework lies in enabling cross-system comparison. If amplitude reflects representational capacity, we can compare capacity across brain areas, individuals, species, or computational models. If steepness reflects integration efficiency, we can compare efficiency across developmental stages, learning states, or clinical populations.

Several limitations should be noted. First, the interpretive mappings are currently theoretical; empirical confirmation is required. Second, the framework assumes sigmoid parameterization, which may not hold for all systems. Third, the relationship between parameters and cognition may be more complex than the simple mappings proposed here.

Future directions include empirical testing of the predictions outlined above, refinement of interpretative mappings based on empirical results, and extension to additional parameterization forms.

This framework establishes that dimensionality analysis can support hypothesis-driven research in cognitive neuroscience. The formal mapping between multiscale dimensionality structure and interpretable cognitive variables enables a new approach to understanding the neural basis of cognition.

8 Conclusion

We have introduced an interpretive framework that maps multiscale dimensionality parameters to cognitively meaningful variables. Amplitude reflects representational capacity, midpoint reflects organizational integration scale, and steepness reflects integration efficiency. This framework enables testable predictions about how cognitive manipulations should affect dimensionality structure. We outline empirical test pathways using fMRI, EEG/MEG, and behavioral paradigms. This work defines a formal mapping between multiscale dimensionality structure and interpretable cognitive variables, enabling hypothesis-driven applications of dimensionality analysis in neuroscience.

Acknowledgments

The author thanks the Sentient-Field Initiative for discussion and feedback.

References

[1] Mikhail Belkin and Partha Niyogi. Laplacian eigenmaps for dimensionality reduction and data representation. *Neural Computation*, 15(6):1373–1396, 2003.

[2] Elena Facco, Maria d'Errico, Alessandro Rodriguez, and Alessandro Laio. Estimating the intrinsic dimension of high-dimensional data sets. *Scientific Reports*, 7(1):12140, 2017.

[3] Juan A Gallego, Matthew G Perich, Lee E Miller, and Sara A Solla. Neural manifolds in the motor system. *Trends in Neurosciences*, 40(4):244–254, 2017.

[4] Elizaveta Levina and Peter J Bickel. Maximum likelihood estimation of intrinsic dimension. *Advances in Neural Information Processing Systems*, 17:777–784, 2005.

[5] Carsen Stringer, Marius Pachitariu, Nicholas Steinmetz, Matteo Carandini, and Kenneth D Harris. High-dimensional geometry of population responses in visual cortex. *Nature*, 571(7766): 361–365, 2019.

[6] Joshua B Tenenbaum, Vin de Silva, and John C Langford. A global geometric framework for nonlinear dimensionality reduction. *Science*, 290(5500):2319–2323, 2000.

[7] A Travesset and D White. Extended singular value decomposition for the analysis of complex datasets. 2022.